

Assignment 2: Recap of GR and TOV Equations

to be handed in online by Sunday, May 10th

Students taking the “Classical and Quantum Aspects of Black Holes” physics course are only expected to submit the exercises marked with an asterisk*, but are obviously invited to try all of them. Students taking the “Analysis and Geometry of Classical Systems” maths course are expected to submit all exercises.

If you decide to skip an exercise, be sure to read it at least, since you will possibly need its result for some later exercise.

We have hidden most hints as white text, to encourage you to try to solve the problems without looking at them. If you are stuck, simply copy the white text somewhere else to reveal it.

Feel free to use `Mathematica` throughout the exercise sheet as long as you submit the notebook or screenshots of it.

Exercise 1: Cosmology II - The Friedmann Equations

Recall that in the previous exercise sheet you studied the flat FLRW metric,

$$ds^2 = -dt^2 + a^2(t) (dx^2 + dy^2 + dz^2) , \quad (1)$$

which you found had

$$R_{tt} = -3\frac{\ddot{a}}{a}, \quad R_{ii} = 2\dot{a}^2 + a\ddot{a}, \quad (2)$$

with $i = x, y, z$ and $\dot{} = \partial_t$.

a)* We now want to study a Universe filled with a perfect fluid,

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}, \quad (3)$$

where U^μ is the fluid's 4-velocity, $U^\mu U_\mu = -1$.

Argue that, if we want to keep homogeneity and isometry, the fluid must be at rest with the FLRW geometry, $U^\mu = (1, 0, 0, 0)$. Show that this implies

$$T^\mu{}_\nu = \text{diag}(-\rho, p, p, p). \quad (4)$$

b)* Using the Einstein Equations and the conservation of the stress-energy tensor $\nabla^\mu T_{\mu\nu} = 0$, derive the Friedmann Equations for a Universe filled with a perfect fluid,

$$(F1) : \quad \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad (5)$$

$$(F2) : \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p), \quad (6)$$

$$(F3) : \quad \dot{\rho} = -3\frac{\dot{a}}{a}(\rho + p). \quad (7)$$

Argue that, if $\rho > 0$ and $p \geq 0$, there cannot be a static solution to the Friedmann Equations.

c)* Assuming that the fluid has a constant equation of state

$$w = p/\rho, \quad (8)$$

solve the Friedmann Equations, that is, find $\rho(a)$ and $a(t)$.

d)* Analyze your solution for the three most common values for w : $w = 0$ for pressure-less matter, such as dust or cold dark matter (CDM); $w = 1/3$ for radiation, such as photons and ultra-relativistic particles like neutrinos; and $w = -1$ for dark energy¹ (this can also be modelled by adding a cosmological constant Λ to the Einstein Equations).

What happens to $a(t)$ when $w = -1$?

Do these three kinds of fluids fail to satisfy any energy condition?

Hint:

Exercise 2: Cosmology III - Slow-roll Inflation

Numerous astronomical observations suggest that the Universe underwent a phase of accelerated ($\ddot{a} > 0$) expansion ($\dot{a} > 0$), called *inflation*, at its birth.

a)* Show that inflation is equivalent to requiring

$$-\dot{H}/H^2 < 1, \quad (9)$$

where $H = \dot{a}/a$ is the Hubble parameter.

Argue also that inflation can be modelled using a perfect fluid with

$$w < -1/3. \quad (10)$$

What energy condition do we need to violate achieve this?

b)* We now model inflation using a scalar field ϕ (the *inflaton*) with potential $V(\phi)$. We still assume homogeneity and isotropy, so the inflaton field can only be a function of time $\phi = \phi(t)$.

Identify the scalar field's energy density ρ_ϕ and pressure p_ϕ

$$\rho_\phi := -T_0^0, \quad (11)$$

$$p_\phi \delta_j^i := T_j^i, \quad (12)$$

from your results for the $T_{\mu\nu}$ of a scalar field from the previous exercise sheet.

Then, inserting these into the first two Friedmann equations, show

$$(F1) : \quad H^2 = \frac{8\pi G}{3} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right), \quad (13)$$

$$(F2) : \quad \dot{H} = -4\pi G \dot{\phi}^2. \quad (14)$$

Finally prove that the Klein-Gordon equation $\square\phi - V'(\phi) = 0$ yields

$$\ddot{\phi} + 3\frac{\dot{a}}{a}\dot{\phi} = -V'(\phi). \quad (15)$$

c)* Argue that, for inflation to happen, we need the potential energy V of the scalar field to be greater than its kinetic energy $\dot{\phi}^2$.

Hint:

¹Recently, the DESI observatory published new results that suggest that dark energy may not have a constant w but have a dynamical one, which may have been $w < -1$ for some time in the past.

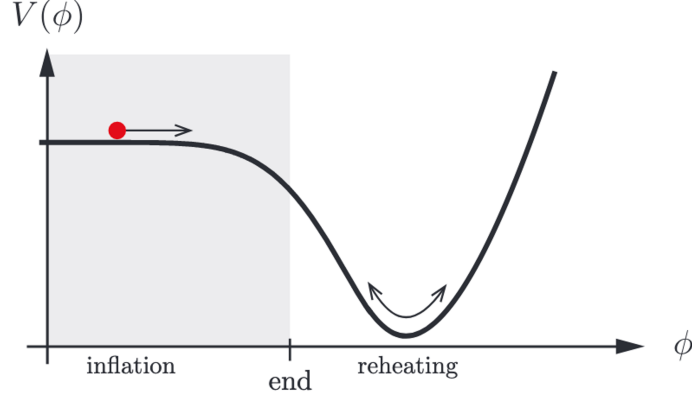


Figure 1: A typical inflationary potential. Inflation happens while the field *slowly rolls* on the almost flat part of the potential, ensuring $V(\phi) > \dot{\phi}^2$.

Exercise 3: Cosmology IV - Einstein's Biggest Blunder?

e) The FLRW metric can be generalized to

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right] = -dt^2 + a^2(t) [d\chi^2 + f_k^2(\chi) d\Omega^2] , \quad (16)$$

where the parameter k describes the curvature of the spatial slices, and

$$f_k(\chi) = \begin{cases} \sin \chi & \text{for } k = 1, \\ \chi & \text{for } k = 0, \\ \sinh \chi & \text{for } k = -1. \end{cases} \quad (17)$$

If $k > 0$, the Universe is positively curved (a closed 3-sphere), if $k = 0$ the Universe is flat, and if $k < 0$ the Universe is negatively curved (an open 3-hyperboloid).

Moreover, we can add a cosmological constant Λ to the Einstein Equations. Then, the new Friedmann Equations for a Universe filled with a perfect fluid read

$$(F1) : \quad \left(\frac{\dot{a}}{a} \right)^2 = \frac{\Lambda}{3} + \frac{8\pi G}{3} \rho - \frac{k}{a^2} , \quad (18)$$

$$(F2) : \quad \frac{\ddot{a}}{a} = \frac{\Lambda}{3} - \frac{4\pi G}{3} (\rho + 3p) , \quad (19)$$

$$(F3) : \quad \dot{\rho} = -3 \frac{\dot{a}}{a} (\rho + p) . \quad (20)$$

Feel free to rederive these results using `diffgeo` as an extra exercise.

Now, assuming that the Universe is filled only with pressure-less matter ($w = 0$), show that there exists a *static* solution ($\ddot{a} = 0$ and $\dot{a} = 0$) to the Friedmann Equations, with a critical density of exactly

$$\rho_c = \frac{\Lambda}{4\pi G} . \quad (21)$$

What must be the curvature of this Universe?

f) Finally, perturb the previous solution $\rho = \rho_c + \delta\rho$, with $\delta\rho \ll \rho_c$, to show that this solution is in fact unstable: a small perturbation $\delta\rho$ induces a perturbation δa that will exponentially grow in time. Einstein's fix did not work, the Universe *must* be dynamical.

Exercise 4: TOV Inequalities

a) Use the Tolman-Oppenheimer-Volkoff equations for dm/dr and $d\Phi/dr$ to eliminate ρ and p from the equation for dp/dr . Show that the resulting equation can be written as

$$(r^{-1}e^{-\Psi}\zeta')' = (mr^{-3})' e^{\Psi}\zeta, \quad (22)$$

where

$$e^{2\Psi} := \left(1 - \frac{2m}{r}\right)^{-1}, \quad \zeta := e^{\Phi}, \quad (23)$$

and the primes denote d/dr derivatives.

b) Use $d\rho/dr \leq 0$ to argue that $(mr^{-3})' \leq 0$.

Hint:

c) It follows that $(r^{-1}e^{-\Psi}\zeta')' \leq 0$. By integrating this equation from r_1 to r and using the equation for $d\Phi/dr$, show that, for any $r_1 \leq r$,

$$\zeta'(r_1) \geq \left(1 - \frac{2m(r)}{r}\right)^{-1/2} \left(\frac{m(r)}{r^3} + 4\pi p(r)\right) \zeta(r) r_1 \left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2}. \quad (24)$$

d) Now, $(mr^{-3})' \leq 0$ implies $m(r_1)/r_1 \geq m(r)r_1^2/r^3$ for $r_1 \leq r$. Use this to show

$$\int_0^r r_1 \left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2} dr_1 \geq \frac{r^3}{2m(r)} \left[1 - \left(1 - \frac{2m(r)}{r}\right)^{1/2}\right]. \quad (25)$$

e) Deduce that

$$\zeta(0) \leq \zeta(r) \left\{1 - \left(\frac{1}{2} + \frac{2\pi r^3 p(r)}{m(r)}\right) \left[\left(1 - \frac{2m(r)}{r}\right)^{-1/2} - 1\right]\right\}. \quad (26)$$

f) By definition $\zeta(r)$ is positive everywhere. In particular, $\zeta(0) > 0$. Deduce that

$$\frac{m(r)}{r} \leq \frac{2}{9} \left(1 - 6\pi r^2 p(r) + \sqrt{1 + 6\pi r^2 p(r)}\right). \quad (27)$$

g) Find *Buchdahl's inequality*,

$$\frac{M}{R} < \frac{4}{9}, \quad (28)$$

where M and R are the total energy and radius of the star.

Exercise 5: Solving TOV for constant density stars

A greatly simplified but instructive solution to the TOV equations is found by assuming

$$\rho(r) = \begin{cases} \rho_0 & r < R, \\ 0 & r > R, \end{cases} \quad (29)$$

where $\rho_0 > 0$ is some positive constant and R is the radius of the star. That is, we are considering a star made from incompressible matter.

a)* Integrate TOV1 to find $m(r)$. Identify the total energy of the star M .

b)* Define $x := p/\rho_0$ to write TOV3 as

$$\frac{dx}{dr} = -(1+x)(1+3x) \frac{4\pi\rho_0 r}{3-8\pi\rho_0 r^2}. \quad (30)$$

c)* Integrate the previous ODE while imposing $p(R) = 0$ and $p(0) = p_c$ to find

$$p_c = \rho_0 \frac{1 - \sqrt{1 - 2M/R}}{3\sqrt{1 - 2M/R} - 1}. \quad (31)$$

d)* Impose $p_c < \infty$ to find Buchdahl's inequality for constant density stars,

$$\frac{M}{R} < \frac{4}{9}. \quad (32)$$

Show that constant density stars can get arbitrarily close to saturating this bound.